Conditioning a Frozen Language Model on PLaMo Translation Hidden States: An Exploratory Study

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Abstract

We test a learned connection from PLaMo 3 Translate 0.8B hidden states to Qwen2.5-1.5B-Instruct while freezing both pretrained backbones. An affine projection converts pooled source states into a target soft prefix and is fitted on 1,024 Japanese arithmetic questions. Across three initializations, the main bridge achieves 4.3% mean synthetic test accuracy, compared with 53.1% for generated English translation supplied as ordinary text and 73.8% for a learned textual prefix with far fewer parameters; these figures use a last-number diagnostic that tolerates explanations. The bridge reaches 1.6% on Japanese MGSM. Source shuffling lowers the seed-11 synthetic score, but the gain over an untrained interface is inconclusive under the numeric diagnostic. A separate 16-question resubstitution experiment confirms that the bridge can memorize source-dependent answers, while its main fit also has low training accuracy. Raw predictions, strict-format scores, decoding-cap counts, layer and prefix ablations, paired intervals and reproducible code document this negative result. For the tested configuration, preserving question text is substantially more effective than injecting translation hidden states without further alignment.

1 Question and scope

Can a translation-specialized language model provide a useful continuous input to a separate, frozen instruction-following language model? The usual translation pipeline answers a Japanese question by first generating an English translation and then passing that text to an answering model. Here, the alternative interface passes pooled hidden states from the translation model through a learned linear projection into the answering model's embedding space. The source model does not generate an English translation on this path. Both pretrained backbones remain fixed.

This interface could avoid committing to a single translated string, but that motivation does not establish an advantage. A hidden-state sequence is neither a lossless encoding of the input nor an automatically compatible representation for another model. Pooling discards information, and a causal decoder's early states cannot represent words that appear later in its input. The projection must also learn an interface that the target model can use. The empirical question is therefore whether this particular trained connection supports accurate free-running answers, and how its behavior compares with ordinary textual inputs.

The architectural precedent is LangBridge, which maps multilingual encoder states into a frozen language model through a learned interface (Yoon et al., 2024). LangBridge studies cross-lingual transfer after alignment using English data. The present study instead uses a frozen translation-specialized causal decoder, pooled intermediate states, and Japanese task-supervised fitting. It is an exploratory adaptation of that general approach, rather than a reproduction of LangBridge or a new claim of zero-shot multilingual alignment.

The source is PLaMo 3 Translate 0.8B (Preferred Networks, 2026), and the target is Qwen2.5-1.5B-Instruct (Qwen, 2024; Qwen et al., 2024). Controlled Japanese arithmetic questions provide exact references and explicit train, validation and test partitions. A separate template holdout tests sensitivity to surface wording. The Japanese MGSM benchmark provides a distinct natural-language endpoint (Shi et al., 2022); its evaluation is reported separately from the generated tasks. These endpoints address the behavior of a small, task-supervised system, not broad multilingual competence.

2 Comparison logic

The practical baselines give the frozen target a Japanese question, its paired English version, or an English translation generated by PLaMo. The paired English version is a reference-language diagnostic rather than a deployable translation system. The bridge receives task supervision that these frozen textual baselines do not. Consequently, their comparison alone cannot isolate a benefit of continuous representations. A learned soft-prefix baseline uses the same supervised training items while retaining the Japanese question as target-model text; it has fewer trainable parameters and is not parameter matched.

Layer and prefix-length ablations test specific interface choices. An untrained projection tests the need for fitting the interface, while source-feature shuffling tests whether a trained system depends on the correspondence between each question and its prefix. Neither control establishes where an arithmetic computation occurs. In particular, source dependence does not by itself show that the target performs reasoning rather than using task-relevant information already present in the source states.

3 Interface and learning objective

Let q be a Japanese question and let $P(q)$ be its official PLaMo translation-request prompt, with Japanese as the source language and English as the requested output language. The prompt ends at the start of the translation output; it contains neither a reference translation nor an answer. At source layer ℓ, a forward pass gives states $H^{(\ell)} \in \mathbb{R}^{T \times d_s}$ for the T valid prompt tokens. The selected source layers are 4, 8 and 16, with layer 8 used for the main configuration. PLaMo's causal attention means that the final valid state is computed after the complete question, while earlier states have access only to preceding input.

Ordered contiguous bins compress these states to a prefix of length K. For bin $j = 0, \ldots, K-1$, define

$$a_j = \left\lfloor \frac{jT}{K} \right\rfloor, \qquad b_j = \left\lfloor \frac{(j+1)T}{K} \right\rfloor, \qquad \bar{h}_j = \frac{1}{b_j - a_j} \sum_{t=a_j}^{b_j-1} H_t^{(\ell)}. \tag{1}$$

Equation 1 applies when $T \geq K$, as in the experimental prompts. The implementation repeats the corresponding valid state for an empty bin when $T < K$. Padding is removed before binning; translation-format tokens remain included. The last bin includes the final valid state. The main prefix has $K = 16$, with $K = 4$ as an ablation.

A shared affine map produces target-space embeddings:

$$z_j = W\bar{h}_j + b, \qquad W \in \mathbb{R}^{d_t \times d_s}, \quad b \in \mathbb{R}^{d_t}. \tag{2}$$

Both models have hidden size 1,536 in the selected configurations, so this bridge has 2,360,832 trainable parameters. Equal dimensions do not make their representation spaces interchangeable; W and b are learned. The projected prefix replaces a temporary marker inside the target chat template's user message. The rest of the prompt asks for the final numeric answer, and the normal assistant-generation boundary follows it. The marker itself is not embedded by the target model. The target receives no Japanese question tokens on the bridge path.

For answer tokens $y_1, \ldots, y_M$, including the termination token, training minimizes

$$\mathcal{L}(W, b) = -\frac{1}{M} \sum_{m=1}^{M} \log p_\theta(y_m \mid U_1, Z, U_2, y_{<m}), \tag{3}$$

where U_1 and U_2 are the fixed template and instruction segments, $Z = (z_0, \ldots, z_{K-1})$, and θ denotes the frozen target parameters. Prompt and padding positions are excluded from the loss. The target forward pass remains differentiable so that gradients reach the bridge through the frozen target. Source states are cached without gradients. At evaluation, greedy decoding starts at the same assistant boundary without answer tokens or teacher forcing.

The supervised textual baseline replaces Z with K learned embeddings shared across questions and keeps the Japanese question in ordinary target text. At $K = 16$ it has 24,576 trainable parameters. It therefore matches the training examples but neither the parameter count nor the information path of the bridge.

4 Experimental design

4.1 Generated questions and partitioning

The controlled dataset contains paired Japanese and English questions for four equally frequent families: addition, subtraction, multiplication, and a two-step operation. The two-step family combines either addition followed by subtraction or multiplication followed by subtraction. The generator computes integer reference answers directly from the sampled operands. Names and objects come from small fixed vocabularies shared across partitions. Each family has two main surface templates; separate templates provide the wording holdout.

The fixed generator seed is 20261108. There are 1,024 training, 128 validation, 256 test and 128 wording-holdout questions, with equal numbers from each family in each partition. Numeric operation-and-operand keys are disjoint across all four partitions. Commutative operand pairs are sorted in these keys, including the first two operands of two-step questions. This prevents a swapped multiplicative pair from appearing as the same arithmetic instance in both training and evaluation. It does not make the arithmetic rules, individual numbers, names or objects unseen. The wording holdout changes both the template and the numeric instance; it is not a paired paraphrase of the ordinary test set.

Before full runs, a small wiring pilot exposed four examples from each preliminary test partition. That pilot revealed a generator key defect and a scoring distinction between numeric correctness and numeric-only formatting. The corrected generator, both scoring rules and common decoding limits were fixed before full experiments. This remains an exploratory evaluation rather than an untouched confirmatory test. The protocol and generated JSONL files preserve the partition rules and the pilot amendments.

4.2 External endpoint

The external evaluation uses all 250 Japanese MGSM test questions from the official repository (Shi et al., 2022). The loader pins repository revision `452a21ad3dae5668c06ceeac21ff073e1e40f9be`. MGSM answers enter scoring only; neither its questions nor its answers are used for bridge fitting or checkpoint selection. The bridge is therefore evaluated under a substantial change from its short synthetic training distribution. Unlike the original MGSM study, this protocol supplies no few-shot chain-of-thought examples and asks for the final number only. The results should not be compared directly with published chain-of-thought scores obtained under other prompting protocols.

4.3 Fitting and comparisons

The main bridge uses source layer 8 and prefix length 16, with independent training seeds 11, 23 and 37. Layer 4, layer 16 and a four-token prefix are separate ablations at seed 11. All fitted conditions use AdamW with learning rate 10^{-3}, zero weight decay, gradient-norm clipping at 1.0, batch size 16 and ten epochs. The saved checkpoint minimizes validation answer-token cross-entropy across the ten epochs. No test score selects a checkpoint or the main layer. The pretrained backbones run in BF16; the bridge parameters and cached pooled features use FP32. The plotted training loss averages minibatch losses; validation loss is weighted by the number of supervised answer tokens. Changing the bridge prefix length changes the target sequence length, while the shared affine map retains the same parameter count.

The learned textual soft prefix uses the same training examples, optimizer, epoch budget and checkpoint rule, at seed 11. Its smaller parameter count and direct access to Japanese text remain material differences. The untrained bridge uses the main architecture and seed without fitting. The shuffled condition keeps the trained main bridge but permutes source feature sequences across evaluation questions, preserving an explicit mapping for each split. The frozen textual systems receive either Japanese, the paired English template, or an actual PLaMo English translation. The external endpoint has no paired-template English condition.

4.4 Decoding, scoring and uncertainty

Every answering condition uses greedy, free-running decoding with the same instruction to return the final number. The cap is 128 new tokens for the generated tasks and 256 for MGSM. PLaMo translation has caps of 128 and 512 tokens, respectively. Per-item records include the generated text, emitted length, normal termination and cap-hit status. The bridge path does not use generated translations or target answer tokens as inference inputs.

The originally specified strict exact match normalizes surrounding whitespace, special-token text, digit width, commas and a terminal period, and requires the complete remaining output to be the reference integer.

It therefore measures answer-format compliance together with correctness. The secondary last-number exact match extracts the final numeric literal from the generated text and compares its exact decimal value with the integer reference. For example, a correct arithmetic explanation ending in a number with a unit can pass the secondary score while failing the strict score. This diagnostic is not a semantic verifier: an unfinished explanation can end in an operand rather than a final answer. Cap-hit rates and raw predictions make this distinction inspectable.

Overall accuracy is the fraction of correct items; equal family counts make the synthetic overall score equal to the mean of the four family accuracies. Selected differences use 10,000 paired bootstrap resamples of evaluation item IDs, with seed 20261108 and percentile 95% intervals. The same sampled IDs are used on both sides of each comparison. These intervals are conditional on the fitted models and the particular evaluation set; they do not quantify all training variability. The three main training seeds are reported separately and summarized without treating seeds as independent test items. The ablations and their intervals are exploratory, with no multiple-comparison correction.

5 Results

5.1 The latent interface is substantially less accurate than retaining text

Table 1 reports both scoring rules for every primary condition. Across the three main bridge seeds, synthetic test accuracy is $4.3\% \pm 2.0$ percentage points (mean $\pm$ sample standard deviation), compared with 53.1% for PLaMo translation followed by ordinary target input. The main bridge mean is 1.0% on the wording holdout and 1.6% on MGSM. All three main seeds score 4/250 on MGSM; equal aggregate scores do not establish stability beyond these three fits. Layer and prefix changes leave bridge test accuracy between 2.0% and 4.3% at seed 11. None approaches the textual alternatives in this experiment.

Table 1: Primary evaluation accuracy: strict / last-number exact match (%)

Condition	Test ($n = 256$)	Wording holdout ($n = 128$)	MGSM ($n = 250$)
Frozen text: Japanese	10.5 / 39.5	11.7 / 43.0	1.2 / 9.2
Frozen text: paired English	17.6 / 50.8	21.9 / 57.0	—
Frozen text: PLaMo translation	14.8 / 53.1	18.8 / 54.7	0.8 / 25.6
Bridge $L8, K16$, seed 11	4.3 / 4.3	1.6 / 1.6	1.6 / 1.6
Bridge $L8, K16$, seed 23	6.2 / 6.2	0.8 / 0.8	1.6 / 1.6
Bridge $L8, K16$, seed 37	2.3 / 2.3	0.8 / 0.8	1.6 / 1.6
Bridge $L4, K16$, seed 11	2.7 / 2.7	2.3 / 2.3	1.2 / 1.2
Bridge $L16, K16$, seed 11	3.1 / 3.1	0.8 / 0.8	1.6 / 1.6
Bridge $L8, K4$, seed 11	2.0 / 2.0	2.3 / 2.3	1.2 / 1.2
Untrained bridge $L8, K16$	1.6 / 2.0	1.6 / 2.3	0.8 / 1.6
Shuffled bridge $L8, K16$	0.4 / 0.4	1.6 / 1.6	1.2 / 1.2
Learned textual prefix $K16$	73.8 / 73.8	60.2 / 60.2	8.0 / 8.0

L is the source layer and K the prefix length. The untrained and shuffled controls use seed 11, as does the learned textual prefix. The latter retains Japanese question text and receives the same task supervision as the bridge; the three frozen text systems receive no such fitting. Paired English templates are available only for the generated tasks. Values are rounded for display; raw counts and predictions remain in the result files.

The learned textual prefix reaches 189/256 (73.8%) on the synthetic test and 77/128 (60.2%) on the wording holdout, with 24,576 trainable parameters. It is more accurate here than the 2,360,832-parameter bridge despite having fewer trainable parameters. On MGSM it reaches 20/250 (8.0%), so the controlled-task improvement does not establish a general natural-language reasoning benefit. Figure 1 displays the conditions with per-condition uncertainty.

5.2 Output format materially changes the apparent ranking

On MGSM, the main bridge's strict score of 1.6% exceeds the frozen translation pipeline's 0.8%, whereas the last-number diagnostic reverses the comparison to 1.6% versus 25.6%. The strict score alone would therefore give a misleading impression of answer quality. The latent systems almost always emit short numeric answers, consistent with their answer-only training target. The frozen systems often emit explanations despite the numeric-only instruction.

No PLaMo translation reaches its translation cap: all 256 test, 128 wording-holdout and 250 MGSM translations terminate normally. Answer generation has a different pattern. The frozen PLaMo-translation

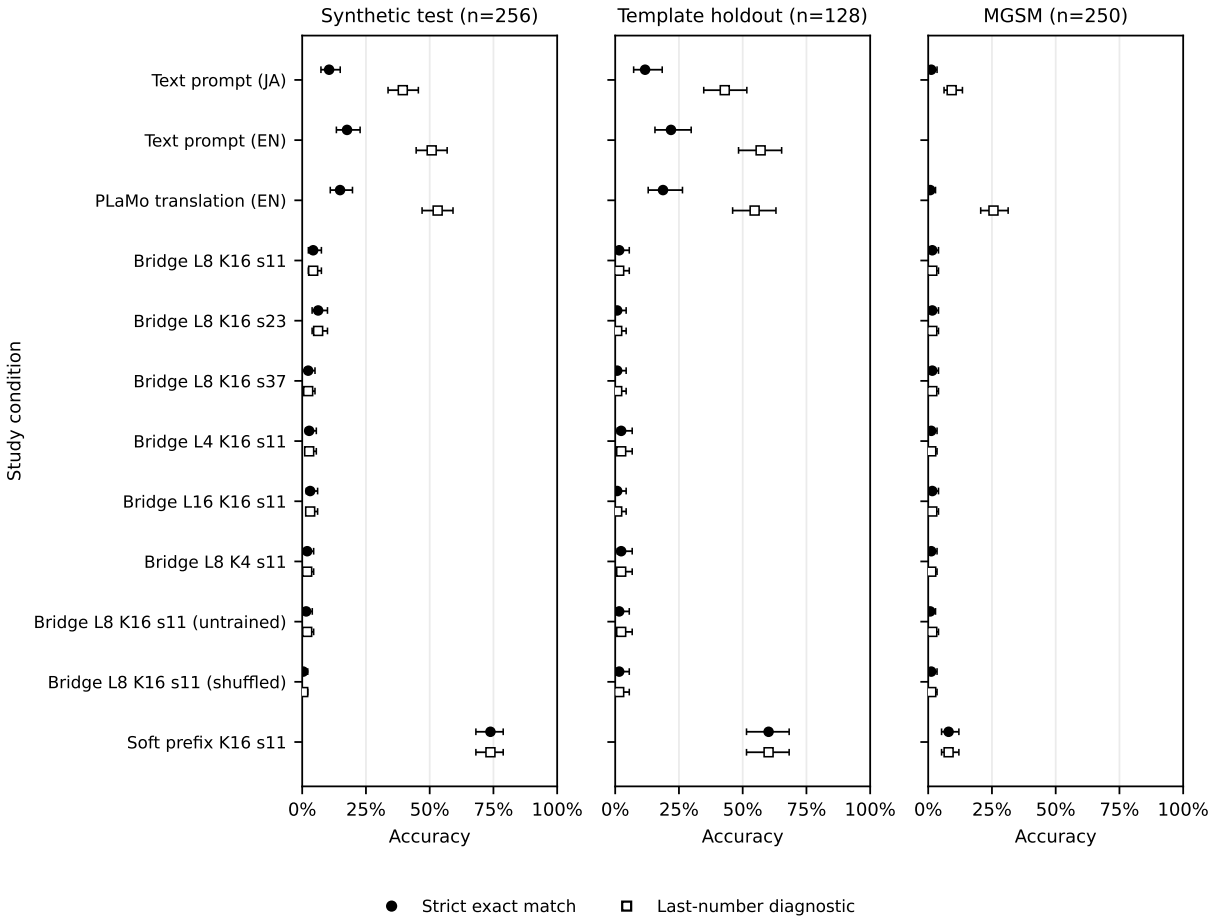


Figure 1: Primary condition accuracies. The panels show synthetic test questions, the separate wording holdout and Japanese MGSM. Marker shapes distinguish strict exact match from the last-number diagnostic. Intervals are 95% Wilson score intervals for per-item binary correctness, conditional on each fitted model. They describe individual proportions; paired effect intervals are reported separately in Table 2. The paired English MGSM condition is unavailable, rather than zero.

pipeline reaches the answer cap on 38/256 synthetic test questions, 26/128 wording-holdout questions and 119/250 MGSM questions. The direct Japanese system reaches it on 1/256, 1/128 and 30/250, respectively. The trained bridges and learned textual prefix have no answer-cap hits on these three endpoints. These results compare finite-budget pipelines; longer explanations and the last-number rule both limit their interpretation as underlying reasoning ability.

5.3 Source correspondence matters on the synthetic test, but the effect is small

For the main seed-11 bridge, shuffling source prefixes reduces synthetic test accuracy from 11/256 to 1/256. The paired difference is 3.9 percentage points, with a 95% bootstrap interval of $[1.6, 6.6]$ points (Table 2). This is evidence that the fitted interface uses source correspondence for some correct answers in this test set. The last-number contrast against the untrained bridge is less decisive: 2.3 points with interval $[-0.4, 5.1]$. On the wording holdout and MGSM, the shuffled contrasts include zero. The result therefore supports limited input dependence rather than broad transfer of reasoning ability.

Table 2: Selected paired contrasts in last-number exact match (percentage points)

Endpoint	Contrast	Difference	95% paired interval
Synthetic test	Bridge − shuffled	$+3.9$	$[1.6, 6.6]$
Synthetic test	Bridge − untrained	$+2.3$	$[-0.4, 5.1]$
Synthetic test	Translation − bridge	$+48.8$	$[42.2, 55.5]$
Synthetic test	Bridge − textual prefix	-69.5	$[-75.4, -63.3]$
Wording holdout	Bridge − shuffled	0.0	$[-3.1, 3.1]$
MGSM	Bridge − shuffled	$+0.4$	$[-1.6, 2.4]$
MGSM	Translation − bridge	$+24.0$	$[18.4, 29.6]$
MGSM	Bridge − textual prefix	-6.4	$[-10.0, -3.2]$

Bridge contrasts use the preselected main $L8, K16$, seed-11 fit. Translation denotes actual PLaMo translation followed by frozen target text input. Intervals use 10,000 paired item resamples; contrast signs follow the displayed subtraction. These are exploratory, conditional intervals without correction for multiple comparisons, rather than equivalence tests.

5.4 The textual prefix improves single-step arithmetic, not all task families

Table 3 shows that the learned textual prefix's high synthetic overall score comes from single-step questions. It answers 183/192 of those items correctly, but only 6/64 two-step items. The paired-English and PLaMo-translation systems perform better on the two-step family under the last-number diagnostic. The learned textual prefix thus does more than change output format on the single-step tasks, while its success remains strongly task dependent.

Table 3: Synthetic test last-number accuracy by task family (%, $n = 64$ per cell)

Condition	Addition	Subtraction	Multiplication	Two-step
Frozen Japanese text	59.4	42.2	43.8	12.5
Frozen paired-English text	65.6	45.3	56.2	35.9
Frozen PLaMo translation	71.9	51.6	57.8	31.2
Main bridge, seed 11	3.1	6.2	1.6	6.2
Shuffled main bridge	1.6	0.0	0.0	0.0
Learned textual prefix	96.9	93.8	95.3	9.4

Figure 2 shows decreasing training and validation losses for the three main bridge fits. Their selected validation losses are 1.205, 1.197 and 1.258, respectively. The learned textual prefix instead selects epoch 9 with validation loss 0.349. The main seed-11 bridge also achieves only 89/1,024 (8.7%) when freely generating answers for its own training questions after fitting. The failure therefore cannot be attributed solely to unseen test questions: the main training run itself has low answer accuracy at the selected budget.

5.5 Post hoc capacity check: a small mapping can be memorized

The low primary scores motivated a separate capacity diagnostic after the main matrix had completed. A newly initialized main-architecture bridge was fitted to 16 generated questions with data seed 20261110,

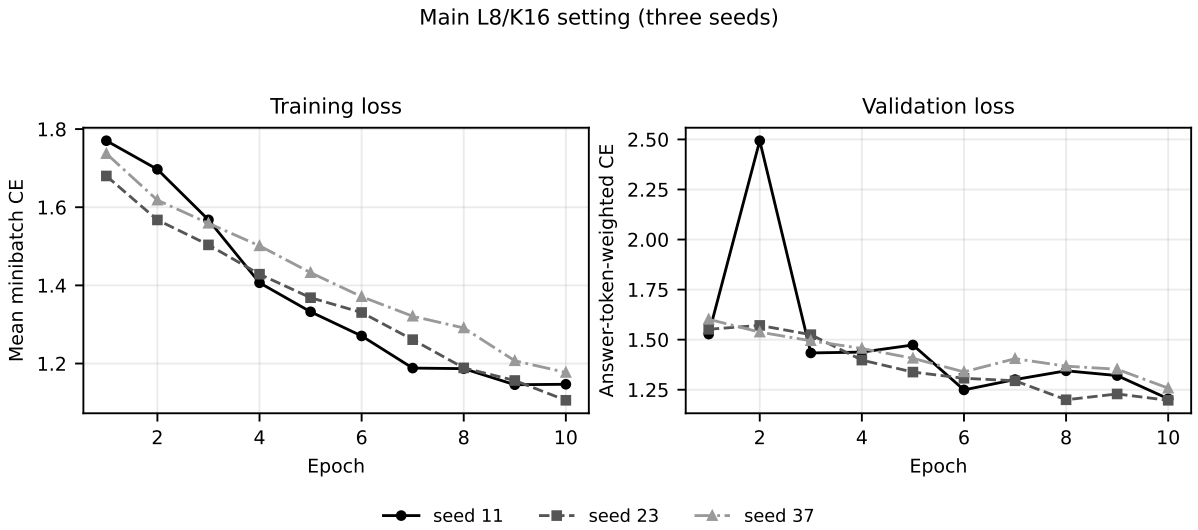


Figure 2: Main $L8, K16$ bridge learning curves for seeds 11, 23 and 37. Training curves average minibatch answer-token cross-entropies; validation curves weight loss by the number of supervised answer tokens. All pretrained backbone parameters remain fixed. Checkpoint selection uses validation loss, while the primary task scores come from independent free-running generation rather than teacher forcing.

batch size 8 and 100 epochs. For this diagnostic only, validation was an exact copy of those training items, and checkpoint selection minimized resubstitution loss. The dataset metadata explicitly marks the intended overlap. This condition measures memorization capacity, not held-out performance, and changes both dataset size and the number of exposures per question.

The fitted bridge answers all 16 training questions correctly. When their source prefixes are shuffled, only 3/16 answers match the original target references, but all 16 outputs equal the answer associated with the *supplied source* question. The three remaining target matches consist of one fixed point in the permutation and two coincident reference answers. Thus the interface can memorize a small source-dependent mapping; its poor main result is not evidence that the computational path is incapable of transmitting any learned correspondence. Together with the main fit's 8.7% training accuracy, the diagnostic points to difficulty fitting the larger problem under the selected representation and optimization budget, rather than a failure confined to unseen questions. It does not identify which architectural or training change would resolve that difficulty.

The diagnostic has its own configuration, data, histories, source-shuffle mapping and raw predictions under `overfit_diagnostic`; its resubstitution scores are excluded from Table 1 and the primary accuracy figure.

6 Interpretation and next experiment

For this model pair and training budget, retaining the question as text is the better-supported practical choice. The translation pipeline is substantially more accurate than the latent bridge under the numeric diagnostic, and a much smaller learned textual prefix performs best on the controlled single-step tasks. The bridge's synthetic source-shuffling effect and small-set memorization show that a learned numerical connection is possible. They do not establish that this interface transfers the translation model's multilingual competence or improves the target's reasoning.

Several differences limit comparison with LangBridge's reported gains. The source here is a causal translation decoder rather than a bidirectional multilingual encoder; its intermediate states are compressed into fixed bins. Alignment uses only 1,024 Japanese question–answer pairs with short numeric targets, rather than a large English alignment corpus. The source states are used at their native scales. A post hoc descriptive calculation on the pooled training features gives RMS values of 11.24, 16.95 and 0.726 for layers 4, 8 and 16, respectively. Layer ablations therefore change both the represented information and its scale. These observations identify differences to investigate, not demonstrated causes of the accuracy gap.

The main limitations are the small generated training distribution, a single source/target model pair, one training seed for each ablation and the exploratory test exposure described in Section 4. Bootstrap

intervals describe item variation conditional on those fits, not broad model or language generalization. MGSM also differs substantially from the generated tasks, and pretraining exposure to benchmark material cannot be ruled out. The output cap and the last-number diagnostic further restrict interpretation of the frozen models' long explanations. Conversely, the learned textual prefix's strong single-step scores do not extend to two-step questions or establish a natural-benchmark improvement.

A useful next experiment is to test whether the same bridge can reconstruct the paired English question from its source states before fitting answer generation. This would provide a direct check of information recovery through the interface and denser alignment supervision. It should be compared with direct answer fitting at a controlled data and compute budget, while preserving an untouched evaluation set. Normalizing native feature scales is a separate, inexpensive control before attributing differences to source-layer depth. Neither modification has been evaluated in the present study.

7 Conclusion

A frozen PLaMo translation model can be connected differentiably to a frozen Qwen model through a small learned projection, and the connection can memorize a tiny source-dependent answer mapping. The tested affine, pooled interface nevertheless has low accuracy on both its larger training set and held-out questions. Ordinary text input, with translation or a learned textual prefix, is markedly more effective in this setting. The experiment supports further interface-alignment diagnostics rather than a claim of improved multilingual reasoning from hidden-state injection alone.

8 Reproduction and recorded settings

The experiment implementation is versioned at Git commit `a2f72d5` (full revision in the execution record). The final plotting changes and isolated capacity-diagnostic script are versioned at `5becef5`; primary training and answer-generation code is unchanged between these commits. The execution record, upstream revisions and measured environment are stored in `results/main/provenance/`. Generated questions, raw predictions, fitted-run histories, selected-epoch metadata and result-driven figures are included in the repository. Upstream weights and source-feature caches are downloaded or generated locally. The public code license does not replace the upstream model or MGSM licenses.

The full experiment is run from the repository root:

```
bash scripts/run_full_experiment.sh
```

The script generates the fixed data, extracts source states, generates actual translations, evaluates the frozen textual systems, trains all fitted conditions, evaluates their held-out outputs and produces the summary and figures. The project README describes environment preparation and individual commands. The separate resubstitution diagnostic is reproduced with `bash scripts/run_overfit_diagnostic.sh`. The isolated smoke command and offline test suite are:

```
.venv/bin/python -m plamo_bridge --config configs/smoke.json \
  smoke --regenerate-data --overwrite
.venv/bin/python -m pytest
```

The implementation was produced with DeepSeek V4.1 Flash. Tests check arithmetic references and canonical split keys, valid-token pooling, prompt and answer boundaries, causal loss alignment, bridge gradients with frozen target parameters, variable-length batched generation, fractional seed averages, and feature/translation cache provenance. A real-model GPU smoke verifies the complete source-to-target path. These checks establish implementation properties; task performance comes from the separate full evaluations reported above.

Table 4 lists the fixed settings and their canonical records. Selected epochs, losses, wall times and peak allocated memory are recorded per run rather than inferred from the nominal configuration. Timing scopes distinguish model loading, source extraction, translation and target answer generation. They describe this implementation on this host, rather than a calibrated hardware benchmark or a general deployment latency claim.

Table 4: Fixed experimental settings and authoritative records

Setting	Value	Record
Source model	PLaMo 3 Translate 0.8B; 16 layers; hidden size 1,536	Model config; feature manifests
Source revision	`2c6192f7` (full revision in configuration)	`configs/default.json`
Target model	Qwen2.5-1.5B-Instruct; 28 layers; hidden size 1,536	Model config; run metadata
Target revision	`989aa798` (full revision in configuration)	`configs/default.json`
Representations	Layers 4, 8, 16; ordered valid-token pooling; $K = 16$ or 4	Feature manifests
Trainable parameters	Affine bridge: 2,360,832; soft prefix: 24,576	Fitted-run metadata
Numerical precision	BF16 pretrained backbones; FP32 projection and pooled cache	Configuration; implementation
Generated partitions	Train 1,024; validation 128; test 256; wording holdout 128	`data/dataset_meta.json`
Generator seed	20261108; disjoint canonical numeric tuples	Dataset metadata
External endpoint	All 250 Japanese MGSM test items; evaluation only	MGSM source provenance
Initialization seeds	Main bridge: 11, 23, 37; other fitted conditions: 11	Fitted-run configuration
Optimizer	AdamW; learning rate 10^{-3}; weight decay 0	Fitted-run configuration
Training budget	Batch 16; ten epochs; gradient clip norm 1.0	Fitted-run configuration
Checkpoint rule	Lowest validation answer-token cross-entropy	History and run metadata
Answer decoding	Greedy; 128 tokens on generated tasks; 256 on MGSM	Prediction metadata
Translation decoding	Greedy; 128 tokens on generated tasks; 512 on MGSM	Translation metadata
Uncertainty	10,000 paired item bootstrap resamples; seed 20261108	Summary JSON
Compute	AMD Radeon RX 9070 XT; 15.922 GiB reported VRAM; six CPU threads	Environment; configuration
Runtime	PyTorch 2.12.0 with ROCm 7.14; Transformers 4.57.1	Environment JSON

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